Repo Download:

Verify Installed: which sshpass git python3 ssh

- which sshpass /usr/bin/sshpass
- which git /usr/bin/git
- which python3 /usr/bin/python3
- which ssh /usr/bin/ssh

If not installed:

- Rocky
  - sudo dnf install -y sshpass
  - sudo dnf install -y git
  - sudo dnf install -y python3
  - sudo dnf install -y openssh-clients
- Verify
  - ssh -V
- Ubuntu/Debian
  - sudo apt update
  - sudo apt install -y sshpass
  - sudo apt install -y git
  - sudo apt install -y python3
  - sudo apt install -y openssh-client
- Verify
  - ssh -V

Download Repo:

- git clone https://github.com/CCDC-Century-College/CCDC-2026.git
- git clone -b beta https://github.com/CCDC-Century-College/CCDC-2026.git

Run Scripts:

- cd CCDC-2026 && ls -la
- python3 linux_internal_credential_rotation.py

Rocky Linux x9.x/Ubuntu x22.x/Debian x12.x Hardening

Step#1 PIDs

- A: Username Check
  - getent passwd <username>
  - id <username>
  - who -u
  - ps -u <username> -f
- B: Red Flags
  - readlink -f /proc/<PID>/exe
  - kill -TERM <PID>
  - passwd -l <username>
  - sleep 3
  - ps -p <PID> -f
  - kill -9 <PID>

Step#2 Network Ports

- A:
  - ss -tulpn (Back to Step#1B with PIDs)

Step#3 Persistence

- A: Cron Jobs/Scheduled Tasks
  - crontab -l
  - sudo crontab -u root -l
  - ls -la /etc/cron* /var/spool/cron/
- B: Red Flags
  - cat /etc/cron.<?>/<filename> (Red Flags continue) (If nothing, continue to Step#4)
  - pgrep -fl <script_name> (Step#1B with PIDs)

Step#4 Persistence systemd

systemctl list-unit-files --state=enabled

- A: States Investigation
  - systemctl cat <servicename> (Red Flags Notes Step#1B) (If nothing, continue to part B)
  - systemctl stop <servicename>
  - systemctl disable <servicename>
  - pgrep -fl <binarynamefromexecstart> (Go back to Step#1B with PIDs)
- B: Activates Column Investigation
  - systemctl list-timers --all (Red Flags found 1.) (NO Red Flags continue)
    1. systemctl cat <activatedservice>
  - systemctl stop <timername>
  - systemctl disable <timername> (Back to A if what?)
- Verify: Timer was disabled.
  - systemctl is-enabled <timername>

Step#5 Firewall/Network Sockets

- A: Firewall Status
  - systemctl status firewalld
  - firewall-cmd --state
  - firewall-cmd --list-all (Red Flags)
    - systemctl enable --now firewalld (If firewall is disabled)
- B: Manual IP Blocking
  - firewall-cmd --permanent --add-rich-rule='rule family="ipv4" source address="<IPaddress>" reject'
  - firewall-cmd --reload
- Verify: IPs are blocked
  - firewall-cmd --list-rich-rules

Ubuntu/Debian

- A: Firewall Status UFW
  - sudo ufw status verbose
  - sudo ufw show added
  - sudo ufw enable (If firewall disabled)
  - sudo ufw logging medium (If logging low)
- B: Manual IP Blocking
  - sudo ufw status numbered
- C: Live Log Monitor
  - sudo tail -F /var/log/ufw.log |fold -s -w $(tput cols)

Quick CLI Commands	Function
w	Active users
who	Logged in users
last -a	Login history
pstree -p	View process parent/child tree
ps -o pid,ppid,user,cmd -p <pid>	Find parent PID and owner
pgrep -af <process>	Locate process and PID
systemctl --failed	Failed services
systemctl --type=service --state=running	Running services
systemctl list-unit-files --type=service	Installed services
systemctl status <service>	Service Health
systemctl is-active <service>	Verify Running
journalctl -u <service>	Service logs
journalctl -xe	Recent system errors

Quick Recovery CLI Commands

Action	Recovery
Lock User	passwd -u <user>
Kill –9 <PID>	systemctl restart <service>
Disable Service	systemctl enable --now <service>
Disable Timer	systemctl enable --now <timer>